

Background & Research Question

Property data is often reported by counties which use differing vocabulary and granularity to describe the same characteristics. How can we detect and normalize these discrepancies to enable downstream analytics?

Data

- Public National Structure Inventory (NSI) data was aggregated to H3 level 9 hexagon cells, yielding 9M+ structure records across 221,108 cells spanning 58 California counties.
 - While the dataset contains additional structural attributes, our analysis focused on structure damage category, building type, and 38 synthetic color labels.
- 13 landcover classes (urban, forest, urban+grass, etc.), sourced from Overture, introduce environmental heterogeneity that may underlie reporting discrepancies.

Exploratory Data Analysis

- Median exposure is 21 structures per H3 cell, but nearly 40% of cells contain fewer than 10 structures. Rural and forest counties are systematically sparse.
- Extreme relative frequencies in low-exposure cells are often statistical artifacts, not true anomalies.
- Color diversity increases almost mechanically with exposure ($r \approx 0.93$). Low-exposure cells appear homogeneous simply due to small sample size.

Figure 1

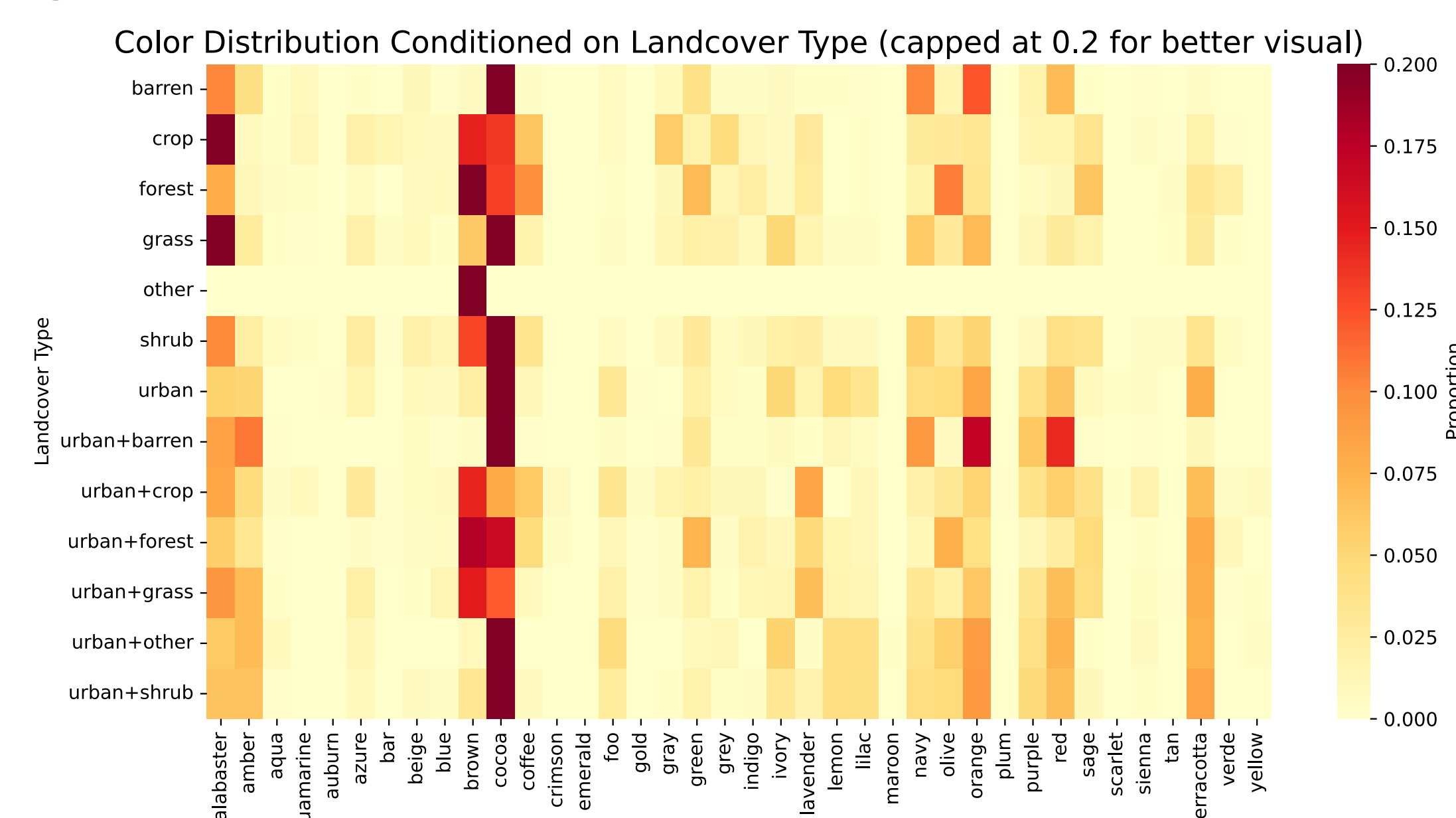


Figure 2A

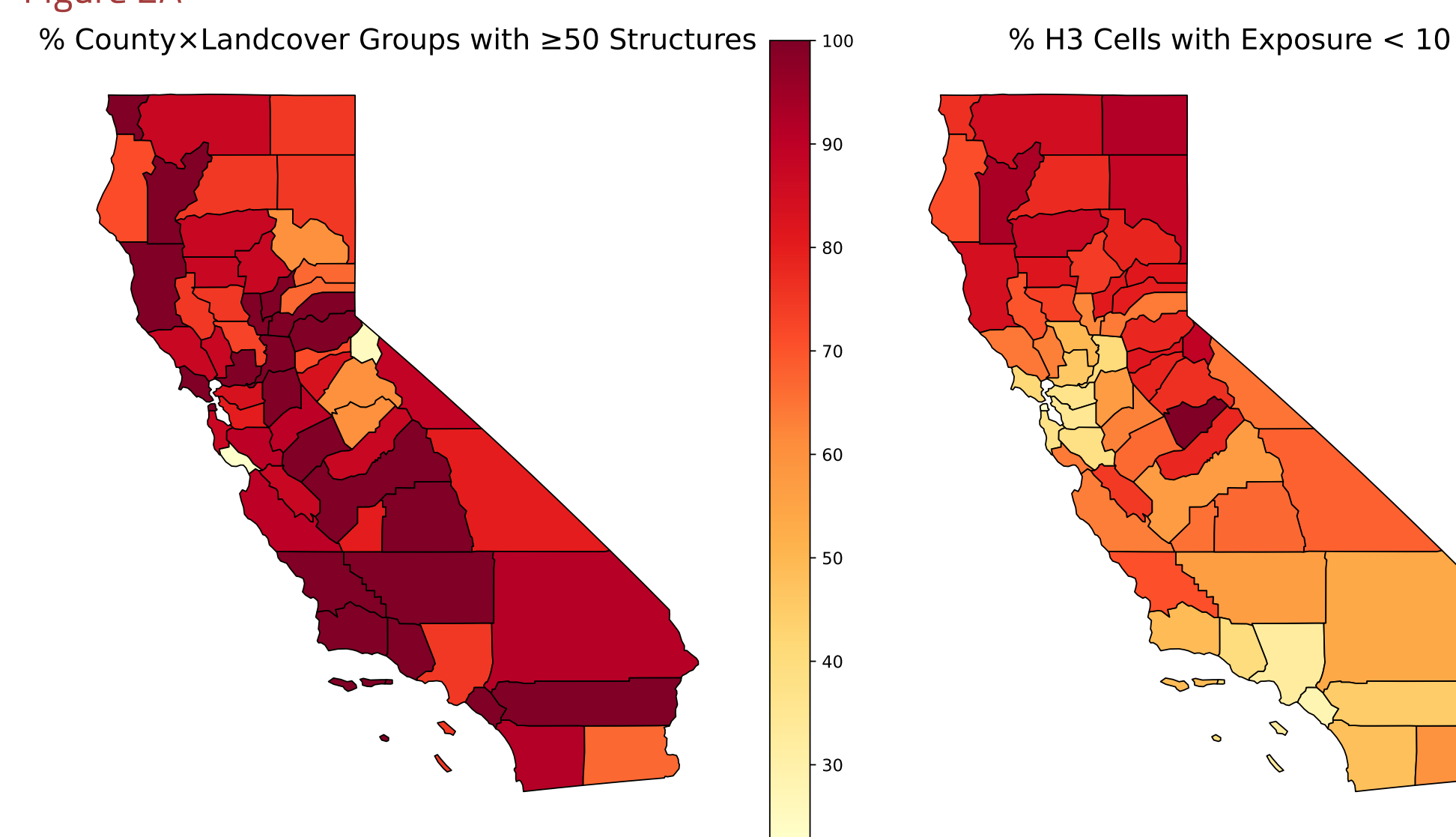
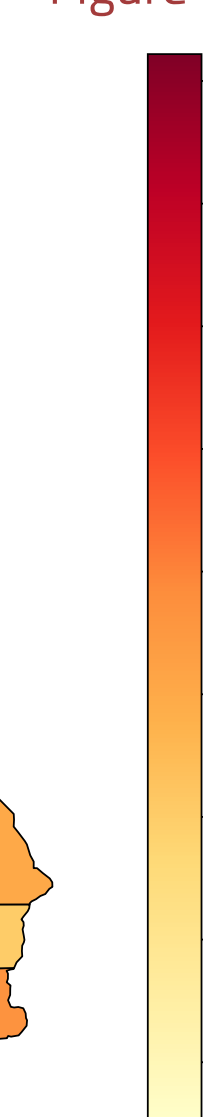


Figure 2B



Method 1: Conditional Probability & Spatial Pooling

- Neighboring counties share similarities (geography, patterns, reporting practices) Measure how likely a color is to show up in a given landcover type
- Mean surprisal (unweighted): 4.17, Exposure-weighted mean surprisal: 2.50

Figure 3

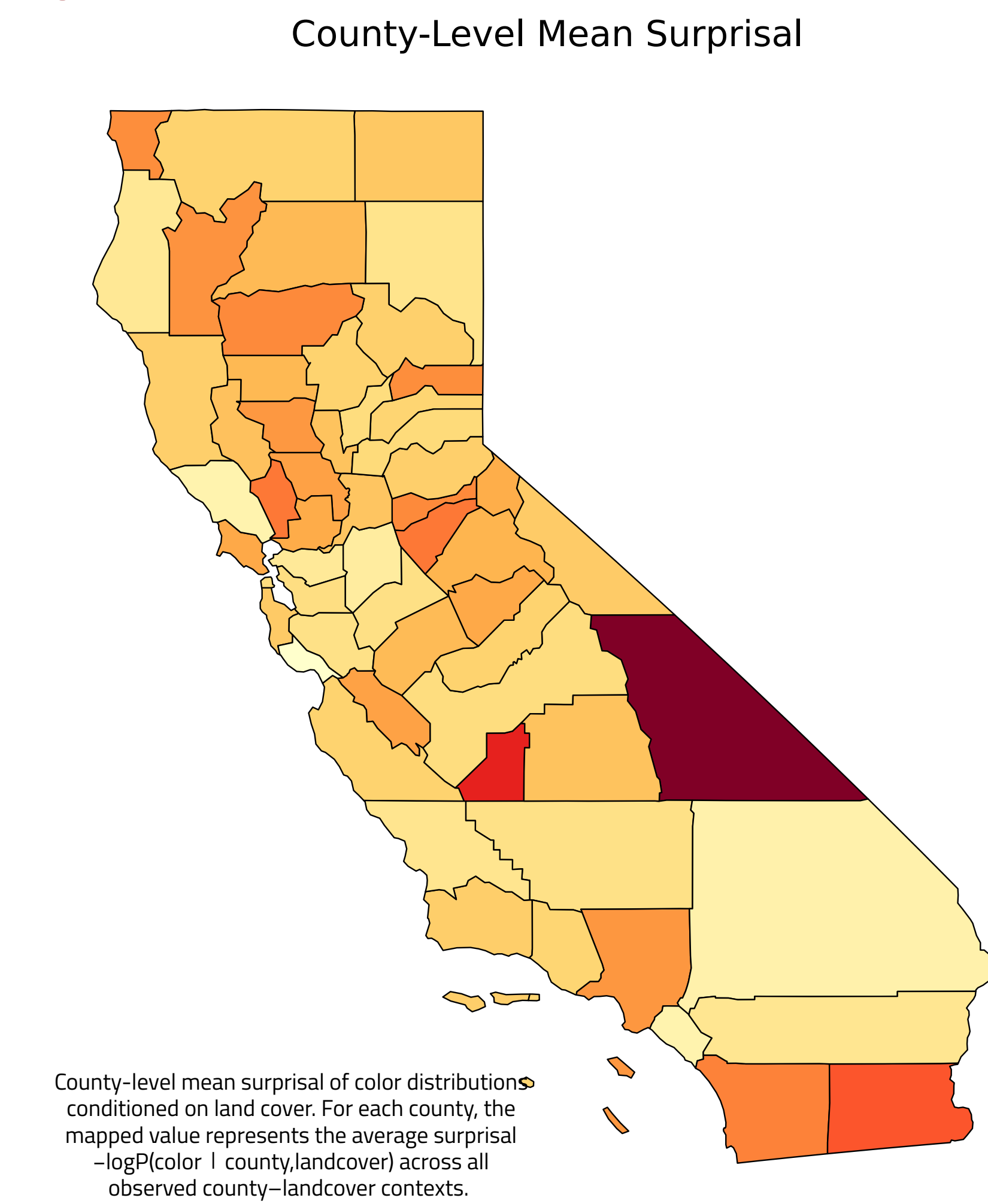


Figure 4

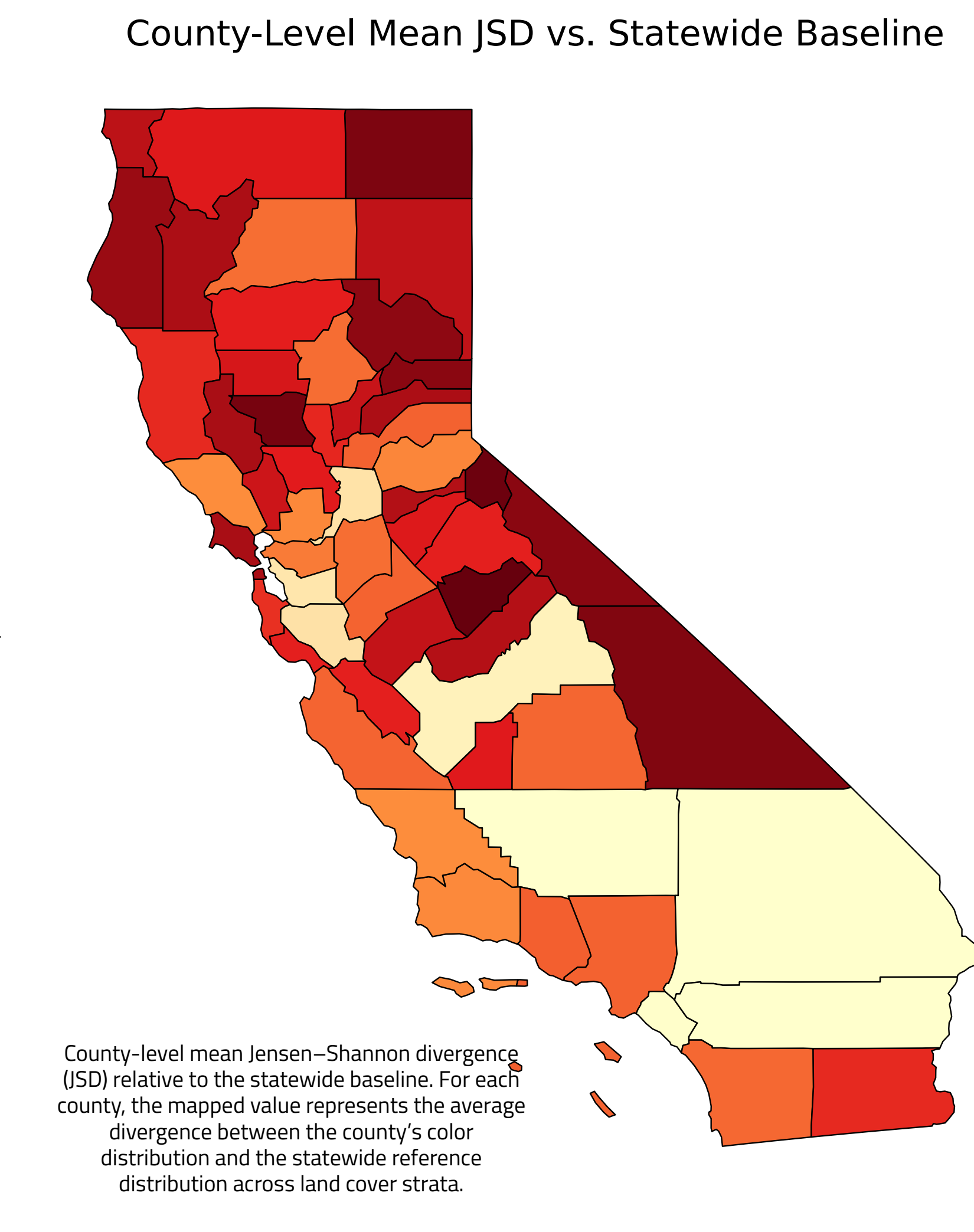
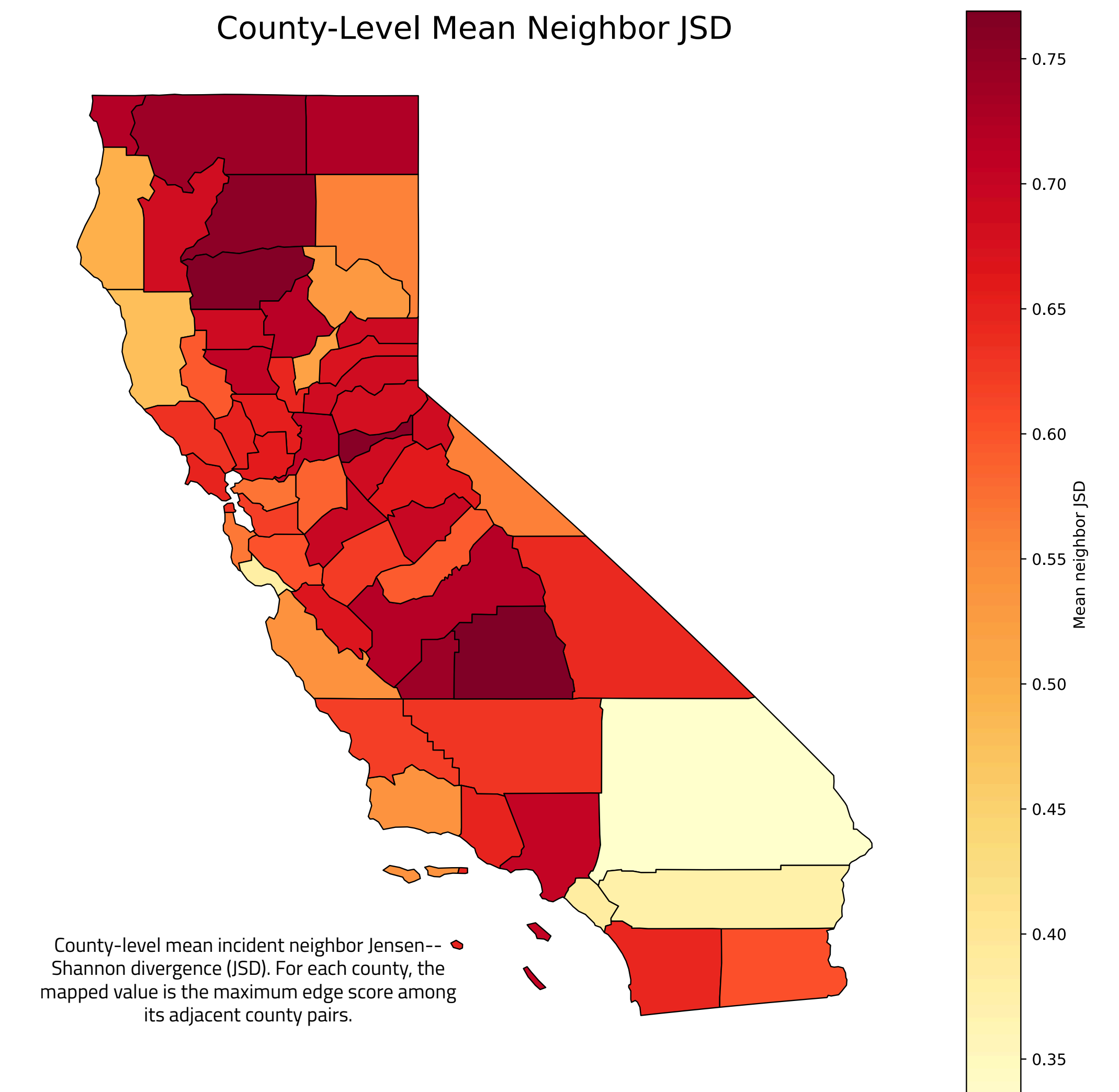


Figure 5



Results

- Through our EDA and method results, we found that neighboring counties would use different color labels to represent the same type of building structure. For example, one county might exclusively use "cocoa" and its neighbor uses only "brown" for the same land cover type, structure damage category, and building type.
- To address this, we built a greedy iterative merging algorithm that identifies color equivalents by voting on which exclusive color pairs appear interchangeably across county strata, merging the lower-frequency label into the higher-frequency one. After 6 rounds, the algorithm converged on 7 color groups (red, navy, cocoa, olive, alabaster, amber, orange) and 6 singletons, which reduced the mean neighbor JSD from **0.62** to **0.216**. One potential outlier is the red group, which absorbed blues and purples due to overlapping structural distributions.
- In conclusion, these results demonstrate that a greedy merging approach can reduce cross-county divergence caused by inconsistent labeling conventions. We found the algorithm to be a good foundation, but future work could improve it by testing clustering-based approaches. This project highlights how inconsistent categorical fields can propagate labeling artifacts into downstream risk assessment models.

Figure 6

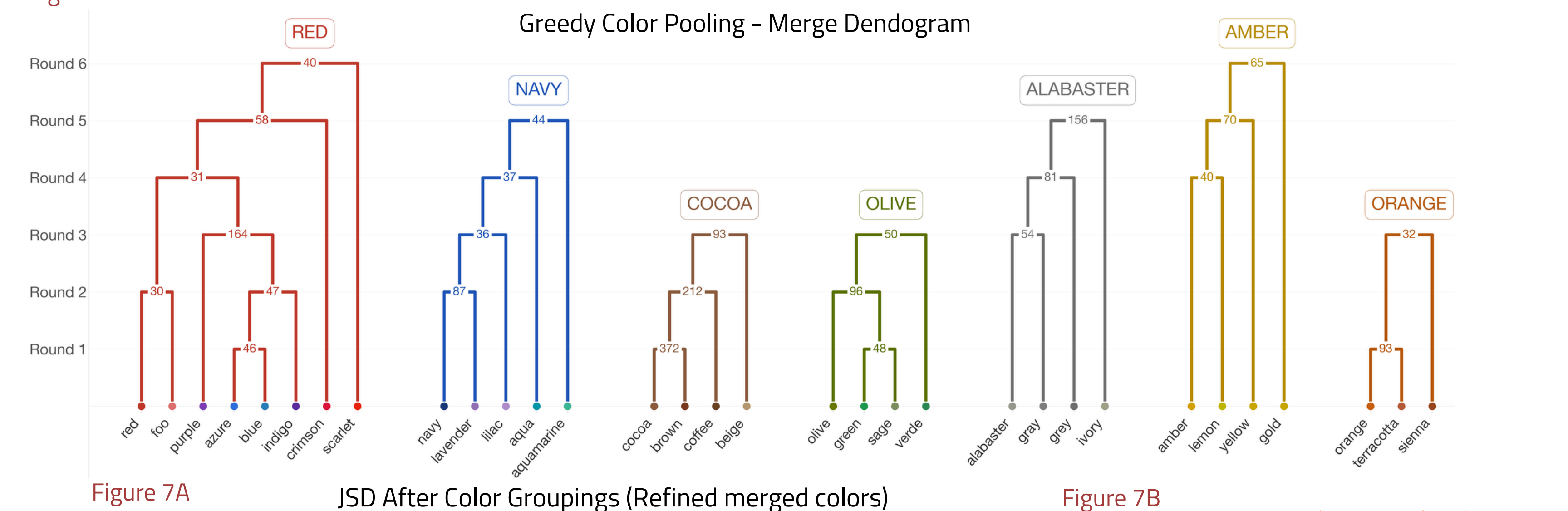
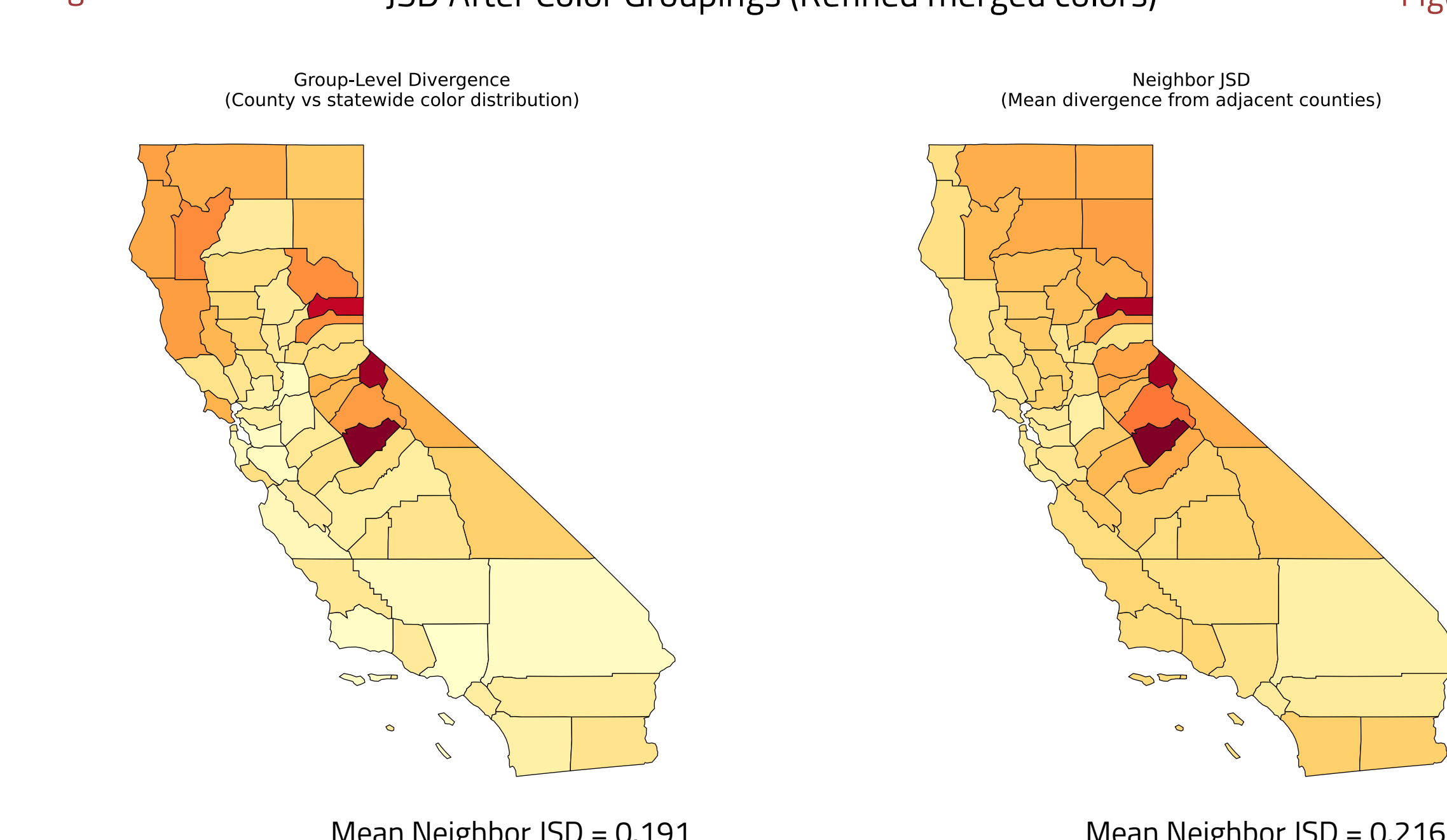


Figure 7A

JSD After Color Groupings (Refined merged colors)

Figure 7B



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